AP Calculus BC
Unit 1 Review Book

Limits and Continuity

This review book is written for a **full beginner-friendly final review** of AP Calculus BC Unit 1. It is detailed enough for students who want a complete reset, but organized so that you can still move quickly. Every major idea is explained in plain language, every important formula is shown in rendered mathematics, and each section includes representative AP-style examples and common traps.

Course Unit:	Unit 1
Scope:	Limits, limit notation, algebraic techniques, continuity, asymptotes, and the Intermediate Value Theorem
Best Use:	Read lesson by lesson, then review the summary sheet and checklist at the end for memorization and speed practice

Prepared for AP Calculus BC review

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How to Use This Book

This book is built for **understanding first, speed second**. For each lesson, keep asking:

1. What is the idea in plain English?
2. What formula, definition, or condition must I know exactly?
3. What does AP usually ask about this topic?
4. What mistake would I be likely to make under time pressure?

A strong Unit 1 student can do all of the following without hesitation:

- distinguish a **function value** from a **limit value**,
- read limits correctly from **graphs, tables, and formulas**,
- simplify indeterminate forms such as $\frac{0}{0}$,
- identify **types of discontinuities**,
- test continuity at a point and on an interval,
- analyze **infinite limits**, **vertical asymptotes**, and **horizontal asymptotes**,
- and use the **Intermediate Value Theorem** with correct reasoning.

1. Unit 1 Big Picture

Unit 1 builds the language of calculus. Before differentiation or integration can make sense, you must understand what it means for a function to *approach* a value. That is the job of limits.

The three anchor ideas of Unit 1 are:

$$\text{limit} = \text{what a function is approaching}$$

$$\text{continuity} = \text{no break between limit and function value}$$

and

$$\text{calculus begins when we study change at an exact instant by using limits.}$$

The most important separation in this unit is the difference between:

$$\lim_{x \to a} f(x) \quad \text{and} \quad f(a).$$

These can be equal, but they do not have to be.

2. 1.1 Introducing Calculus - Can Change Occur at an Instant?

2.1 Core idea

This lesson motivates calculus through rates of change. The key question is: how can we describe change *at one exact moment* rather than over a whole interval?

2.2 Essential formulas

The average rate of change of a function on $[a, b]$ is

$$\frac{f(b) - f(a)}{b - a}.$$

This is the slope of the secant line.

The instantaneous rate of change at $x = a$ is the limit

$$\lim_{h \to 0} \frac{f(a + h) - f(a)}{h}.$$

This is the slope of the tangent line, which later becomes the derivative.

2.3 What AP asks

- Distinguish average rate of change from instantaneous rate of change.
- Use smaller and smaller intervals to estimate a tangent slope.
- Interpret rates of change in words and units.

Common trap: A secant slope is only an approximation. It becomes a tangent slope only in the limit as the second point moves toward the first.

Example 1.1A. Suppose

$$\frac{s(2.1) - s(2)}{0.1} = 5.8, \qquad \frac{s(2.01) - s(2)}{0.01} = 5.98, \qquad \frac{s(2.001) - s(2)}{0.001} = 5.998.$$

Estimate the instantaneous rate of change at $t = 2$.

Solution. These quotients have the form

$$\frac{s(2 + h) - s(2)}{h}.$$

As $h \to 0$, the values approach 6. Therefore the instantaneous rate of change is approximately

$$6.$$

3. 1.2 Defining Limits and Using Limit Notation

3.1 Core idea

The statement

$$\lim_{x \to a} f(x) = L$$

means that as x gets arbitrarily close to a, the function values $f(x)$ get arbitrarily close to L.

3.2 One-sided limits

The left-hand limit is

$$\lim_{x \to a^-} f(x),$$

and the right-hand limit is

$$\lim_{x\to a^+} f(x).$$

The two-sided limit exists if and only if these agree:

$$\lim_{x\to a} f(x) \text{ exists} \iff \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x).$$

3.3 The three things students confuse

- $f(a)$: the actual function value.
- $\lim_{x\to a} f(x)$: the value the function approaches.
- continuity at a: the limit exists and equals the function value.

Common trap: A function can have a limit at $x = a$ even if $f(a)$ does not exist. The limit only depends on nearby values, not necessarily the value at the point itself.

Example 1.2A. Suppose

$$\lim_{x\to 3^-} f(x) = 2, \qquad \lim_{x\to 3^+} f(x) = 2, \qquad f(3) = 5.$$

Find $\lim_{x\to 3} f(x)$.
Solution. The left-hand and right-hand limits agree, so

$$\lim_{x\to 3} f(x) = 2.$$

The fact that $f(3) = 5$ does not change the limit.

4. 1.3 Estimating Limit Values from Graphs

4.1 Core idea

From a graph, the limit is found by following the curve toward the target x-value from the left and from the right.

4.2 Graph-reading strategy

1. Look at where the graph goes as $x \to a^-$.
2. Look at where the graph goes as $x \to a^+$.
3. If both sides approach the same y-value, that is the limit.
4. Then separately check the actual function value, usually shown by a filled dot.

Common trap: The filled dot tells you $f(a)$, not necessarily the limit. The open circle often shows the value the graph is approaching.

Example 1.3A. A graph has an open circle at $(1, 4)$ and a filled point at $(1, 7)$. From both sides, the graph approaches the open circle. What are $\lim_{x\to 1} f(x)$ and $f(1)$?
Solution.

$$\lim_{x\to 1} f(x) = 4, \qquad f(1) = 7.$$

5. 1.4 Estimating Limit Values from Tables

5.1 Core idea

Tables show numerical evidence of what a function is approaching. You read the values from the left and right and look for a trend.

5.2 What to look for

- Do values from both sides get close to the same number?
- Do they grow without bound, suggesting $+\infty$ or $-\infty$?
- Is the trend different from the actual value at the point?

Common trap: A table can strongly suggest a limit, but it does not directly tell you the exact function value at the point unless that row is given.

Example 1.4A. A table shows that as x approaches 2 from the left and right, $f(x)$ approaches values like 6.99, 6.999, 7.001, and 7.01. Estimate the limit.
Solution. The values suggest

$$\lim_{x\to 2} f(x) = 7.$$

6. 1.5 Determining Limits Using Algebraic Properties of Limits

6.1 Core idea

When the individual limits exist, you can combine them using limit laws instead of starting from scratch.

6.2 Essential laws

If

$$\lim_{x\to a} f(x) = L, \qquad \lim_{x\to a} g(x) = M,$$

then:

$$\lim_{x\to a}[f(x) + g(x)] = L + M,$$

$$\lim_{x\to a}[f(x) - g(x)] = L - M,$$

$$\lim_{x\to a}[cf(x)] = cL,$$

$$\lim_{x\to a}[f(x)g(x)] = LM,$$

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M} \quad \text{if } M \neq 0.$$

For powers and roots,

$$\lim_{x\to a}[f(x)]^n = L^n, \qquad \lim_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{L}$$

when these expressions are defined.

6.3 Direct substitution

For polynomials and for rational functions with nonzero denominator at the point, direct substitution usually works.

Example 1.5A. If

$$\lim_{x\to 2} f(x) = 3, \qquad \lim_{x\to 2} g(x) = -4,$$

find

$$\lim_{x\to 2}[2f(x) - 3g(x)].$$

Solution.

$$2(3) - 3(-4) = 6 + 12 = 18.$$

So the limit is

$$18.$$

Common trap: Do not use the quotient law if the denominator limit is zero. That situation needs a different analysis.

7. 1.6 Determining Limits Using Algebraic Manipulation

7.1 Core idea

When direct substitution gives an indeterminate form such as

$$\frac{0}{0},$$

you often simplify the expression first.

7.2 Common techniques

- factor and cancel,
- rationalize with a conjugate,
- combine fractions,
- simplify a complex fraction.

The form

$$\frac{0}{0}$$

means **keep working**. It does *not* mean the limit is zero.

Example 1.6A. Compute

$$\lim_{x\to 3} \frac{x^2-9}{x-3}.$$

Solution. Direct substitution gives 0/0. Factor:

$$\frac{x^2-9}{x-3} = \frac{(x-3)(x+3)}{x-3} = x+3 \quad (x \neq 3).$$

Therefore,

$$\lim_{x\to 3} \frac{x^2-9}{x-3} = \lim_{x\to 3}(x+3) = 6.$$

Example 1.6B. Compute

$$\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}.$$

Solution. Direct substitution gives 0/0. Rationalize:

$$\frac{\sqrt{x}-2}{x-4} \cdot \frac{\sqrt{x}+2}{\sqrt{x}+2} = \frac{x-4}{(x-4)(\sqrt{x}+2)} = \frac{1}{\sqrt{x}+2}.$$

So

$$\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4} = \frac{1}{2+2} = \frac{1}{4}.$$

8. 1.7 Selecting Procedures for Determining Limits

8.1 Core idea

Not every limit should be attacked the same way. A large part of success in Unit 1 is recognizing the most efficient method.

8.2 Decision guide

- If direct substitution works, use it.
- If substitution gives 0/0, simplify algebraically.
- If the function oscillates but is trapped, try the Squeeze Theorem.
- If you are working from a graph or table, reason visually or numerically.
- If you are studying end behavior, compare dominant terms or asymptotes.

Example 1.7A. Choose the best method for

$$\lim_{x\to 3} \frac{x^2 - 5x + 6}{x - 3}.$$

Solution. Direct substitution gives $0/0$, so the best method is factoring:

$$\frac{x^2 - 5x + 6}{x - 3} = \frac{(x-2)(x-3)}{x-3} = x - 2.$$

Then the limit is 1.

9. 1.8 Determining Limits Using the Squeeze Theorem

9.1 Core idea

If one function is trapped between two others that approach the same value, then the trapped function must approach that value too.

9.2 Statement

If

$$g(x) \le f(x) \le h(x)$$

near $x = a$, and

$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L,$$

then

$$\lim_{x\to a} f(x) = L.$$

9.3 Classic bounded oscillation fact

Because

$$-1 \le \sin\left(\frac{1}{x}\right) \le 1,$$

we get, for example,

$$-|x| \le x \sin\left(\frac{1}{x}\right) \le |x|.$$

If $x \to 0$, both outer expressions go to 0, so the middle one also goes to 0.

Example 1.8A. Compute

$$\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right).$$

Solution. Since

$$-1 \le \cos\left(\frac{1}{x}\right) \le 1,$$

we have

$$-x^2 \le x^2 \cos\left(\frac{1}{x}\right) \le x^2.$$

As $x \to 0$, both $-x^2$ and x^2 approach 0. Therefore,

$$\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right) = 0.$$

Common trap: Students see the oscillating term and immediately say "the limit does not exist." That is not always true if the oscillation is multiplied by something going to zero.

10. 1.9 Connecting Multiple Representations of Limits

10.1 Core idea

AP often gives the same limit information in several forms at once: graph, table, formula, and words. You must translate between them.

10.2 What each representation is good for

- **Graph:** left-hand behavior, right-hand behavior, holes, jumps, asymptotes.
- **Table:** numerical evidence of approach.
- **Formula:** exact algebraic computation.
- **Context:** meaning and units.

Example 1.9A. A table suggests $f(x) \to 5$ as $x \to 2$, a graph shows an open circle at $(2, 5)$, and a filled point at $(2, 1)$. What must be true?
Solution.

$$\lim_{x\to 2} f(x) = 5, \qquad f(2) = 1.$$

This is a removable discontinuity.

11. 1.10 Exploring Types of Discontinuities

11.1 Core idea

A discontinuity is a break in the way a function behaves at a point. AP expects you to recognize the main types.

11.2 Main types

- **Removable discontinuity:** the limit exists, but the function value is missing or incorrect.

- **Jump discontinuity:** left-hand and right-hand limits exist but are different.
- **Infinite discontinuity:** the function grows without bound near the point, often producing a vertical asymptote.
- **Oscillating discontinuity:** the function keeps oscillating and never approaches one value.

Example 1.10A. If

$$\lim_{x\to 3^-} f(x) = 2, \qquad \lim_{x\to 3^+} f(x) = 5,$$

what type of discontinuity occurs at $x = 3$?
Solution. Since both one-sided limits exist but are not equal, the discontinuity is a **jump discontinuity**.

12. 1.11 Defining Continuity at a Point

12.1 Core idea

A function is continuous at a point if the function value and the limit value meet perfectly.

12.2 Three conditions for continuity at $x = a$

A function f is continuous at $x = a$ if and only if:

1. $f(a)$ exists,
2. $\lim_{x\to a} f(x)$ exists,
3. $\lim_{x\to a} f(x) = f(a)$.

For AP free response, these three conditions are the cleanest standard explanation of continuity at a point.

Example 1.11A. Let

$$f(x) = \begin{cases} x+1, & x \neq 2, \\ 5, & x = 2. \end{cases}$$

Is f continuous at $x = 2$?
Solution. For $x \neq 2$, the function behaves like $x + 1$, so

$$\lim_{x\to 2} f(x) = 3.$$

But

$$f(2) = 5.$$

Since the limit and function value are not equal, f is **not continuous** at $x = 2$.

13. 1.12 Confirming Continuity over an Interval

13.1 Core idea

To be continuous on an interval, a function must be continuous at every point in that interval.

13.2 Closed interval reminder

On a closed interval $[a, b]$, continuity means:

- right-continuous at a,
- continuous on (a, b),
- left-continuous at b.

13.3 Functions known to be continuous on their domains

- polynomials,
- rational functions,
- exponential functions,
- logarithmic functions,
- trigonometric functions.

They are continuous wherever they are defined.

Example 1.12A. Determine where

$$f(x) = \frac{x+1}{x^2-4}$$

is continuous.

Solution. Rational functions are continuous wherever the denominator is nonzero. Here,

$$x^2 - 4 = (x-2)(x+2),$$

so the denominator is zero at $x = \pm 2$. Therefore f is continuous on

$$(-\infty, -2) \cup (-2, 2) \cup (2, \infty).$$

14. 1.13 Removing Discontinuities

14.1 Core idea

A removable discontinuity can often be fixed by defining the function value to match the limit.

14.2 The standard pattern

If

$$\lim_{x \to a} f(x) = L$$

exists, but the original function is missing at a or has the wrong value there, then the discontinuity is removable. Defining

$$f(a) = L$$

creates continuity.

Example 1.13A. Let

$$f(x) = \begin{cases} \dfrac{x^2-9}{x-3}, & x \neq 3, \\ k, & x = 3. \end{cases}$$

Find the value of k that makes f continuous at $x = 3$.
Solution. Simplify the expression for $x \neq 3$:

$$\frac{x^2-9}{x-3} = \frac{(x-3)(x+3)}{x-3} = x+3.$$

Thus

$$\lim_{x\to 3} f(x) = 6.$$

To make the function continuous, define

$$k = 6.$$

15. 1.14 Connecting Infinite Limits and Vertical Asymptotes

15.1 Core idea

If a function grows without bound near $x = a$, then the line $x = a$ is a vertical asymptote.

15.2 Infinite limits

Examples include

$$\lim_{x\to a^-} f(x) = +\infty, \qquad \lim_{x\to a^+} f(x) = -\infty.$$

Any one-sided or two-sided unbounded behavior can indicate a vertical asymptote.

15.3 Sign analysis matters

For functions like

$$\frac{1}{x-2},$$

you must determine whether the denominator is a tiny positive number or a tiny negative number.

Example 1.14A. Compute

$$\lim_{x\to 1^-} \frac{2}{x-1}.$$

Solution. As $x \to 1^-$, the denominator $x - 1$ is a very small negative number. A positive number divided by a very small negative number approaches

$$-\infty.$$

So

$$\lim_{x\to 1^-} \frac{2}{x-1} = -\infty.$$

16. 1.15 Connecting Limits at Infinity and Horizontal Asymptotes

16.1 Core idea

A horizontal asymptote is determined by what a function does as $x \to \infty$ or $x \to -\infty$.

16.2 Definition

If

$$\lim_{x\to\infty} f(x) = L \quad \text{or} \quad \lim_{x\to-\infty} f(x) = L,$$

then the line

$$y = L$$

is a horizontal asymptote.

16.3 Rational function shortcut

For

$$\frac{p(x)}{q(x)},$$

compare degrees:

- If degree of numerator $<$ degree of denominator, then $y = 0$.
- If degrees are equal, use the ratio of leading coefficients.
- If degree of numerator $>$ degree of denominator, there is no horizontal asymptote.

Example 1.15A. Compute

$$\lim_{x\to\infty} \frac{3x^2 - 1}{5x^2 + 7}.$$

Solution. The degrees are equal, so the limit is the ratio of leading coefficients:

$$\frac{3}{5}.$$

Thus the horizontal asymptote is

$$y = \frac{3}{5}.$$

17. 1.16 Working with the Intermediate Value Theorem

17.1 Core idea

The Intermediate Value Theorem (IVT) guarantees that a continuous function takes every intermediate value between two outputs.

17.2 Statement

If f is continuous on $[a, b]$, and N is any number between $f(a)$ and $f(b)$, then there exists a number $c \in (a, b)$ such that

$$f(c) = N.$$

A common special case is proving that a root exists: if f is continuous on $[a, b]$ and

$$f(a) \cdot f(b) < 0,$$

then there must be some $c \in (a, b)$ such that

$$f(c) = 0.$$

Common trap: IVT proves *existence*, not uniqueness. It tells you there is at least one such point, not that there is exactly one.

Example 1.16A. Suppose f is continuous on $[1, 4]$, with

$$f(1) = -2, \qquad f(4) = 5.$$

What can be guaranteed by IVT?
Solution. Since 0 lies between -2 and 5, there exists some $c \in (1, 4)$ such that

$$f(c) = 0.$$

Example 1.16B. Explain why the polynomial

$$p(x) = x^3 - x - 1$$

has a zero between 1 and 2.
Solution. Polynomials are continuous everywhere. Evaluate the endpoints:

$$p(1) = 1 - 1 - 1 = -1, \qquad p(2) = 8 - 2 - 1 = 5.$$

Since $p(1) < 0$ and $p(2) > 0$, IVT guarantees a number $c \in (1, 2)$ such that

$$p(c) = 0.$$

18. Unit 1 Summary Sheet

18.1 Definitions and formulas to know cold

$$\lim_{x \to a} f(x) = L$$

means $f(x)$ approaches L as x approaches a.

$$\lim_{x \to a} f(x) \text{ exists} \iff \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x).$$

Continuity at $x = a$ requires:

1. $f(a)$ exists,
2. $\lim_{x \to a} f(x)$ exists,
3. $\lim_{x \to a} f(x) = f(a)$.

Squeeze Theorem:

$$g(x) \le f(x) \le h(x), \qquad \lim g(x) = \lim h(x) = L \Rightarrow \lim f(x) = L.$$

Intermediate Value Theorem: If f is continuous on $[a, b]$, then it takes every value between $f(a)$ and $f(b)$.

18.2 Asymptote reminders

- Vertical asymptote at $x = a$: some one-sided or two-sided limit is $\pm\infty$.
- Horizontal asymptote $y = L$: $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$.

18.3 High-frequency AP mistakes checklist

Mistake	Correct idea
Assuming $f(a)$ and $\lim_{x\to a} f(x)$ are automatically the same	They may be different; always check separately.
Saying $0/0$ means the limit is 0	It is an indeterminate form; simplify first.
Using the quotient law when denominator limit is 0	The quotient law requires a nonzero denominator limit.
Calling every discontinuity a "hole"	Distinguish removable, jump, infinite, and oscillating discontinuities.
Forgetting the three conditions for continuity	A function value, a limit, and equality between them are all required.
Using IVT without checking continuity	Continuity on the interval is required.
Claiming IVT proves exactly one solution	IVT proves at least one solution, not uniqueness.

Final Unit 1 Memory Checklist

Before the AP exam, make sure you can do each of these quickly:

- read a two-sided limit from a graph,
- separate left-hand limit, right-hand limit, and function value,
- evaluate direct-substitution limits,
- simplify a $\frac{0}{0}$ expression by factoring or rationalizing,
- classify a discontinuity,
- test continuity at a point,
- identify vertical and horizontal asymptotes,
- and write a clean IVT justification in one or two sentences.